

Logs Validation of Windows Domain Controller and Domain Joined Client using Extraction and Detection SPL in Splunk Enterprise

Summary

This document has steps related to validating logs in Search Head using `_raw` SPL, building Extraction SPL from the `_raw` and developing Detection logic SPL based on the Extracted fields, Normalizing the logs got from both the events (4728, 4732) and Developing a Correlation logic SPL to verify the Privilege Escalation attack with evidence.

1. Validate raw logs

1) Check Domain Escalation logs (4728)

In Search Head UI > Search and Reporting app,

Run:

```
index=win_ad_idx EventCode=4728
| table _raw
```

Look for:

Group Name

Member: Security ID, Account Name

Subject: Account Name

ComputerName

Expected:

Group Name: Domain Admins

Member: Account Name: : <domain_username>

Member: Security ID: <domainuser_sid1>

Confirms: The chosen user was added to Domain Admins group

The screenshot shows the Splunk Enterprise Search Head interface. The search bar contains the query `index=win_ad_idx EventCode=4728 | table _raw`. The search results show 1 of 316 events matched. The selected event is a Windows security audit log (EventCode=4728) from a source named 'Microsoft Windows security auditing'. The log message states: 'Message=A member was added to a security-enabled global group.' The subject of the log is 'admin' (Security ID: S-1-5-21-3021654831-3858487264-1625118876-1120). The member being added is 'CN=Eric Wan,OU=Employees,OU=Users,OU=NovaSec,DC=splunkadmin,DC=local' (Security ID: S-1-5-21-3021654831-3858487264-1625118876-512). The group being added to is 'Domain Admins' (Security ID: S-1-5-21-3021654831-3858487264-1625118876-512).

2) Check Local Escalation logs (4732)
In **Search Head UI > Search and Reporting app**,
Run:

```
index=win_idx EventCode=4732  
| table _raw
```

Look for:

Subject: Account Name
Member: Security ID, Account Name
Group Name
ComputerName

Expected:

Group Name: Administrators
Member: Account Name: (might be empty OR Admin selected
Domain user name)
Member: Security ID: <domainuser_sid1>

Confirms: The chosen user was added to local Administrators group

The screenshot shows the Splunk Search Head UI with the following details:

- Search Query:** `index=win_idx EventCode=4732 | table _raw`
- Results:** 1 event (4/05/2026 10:30:58.000 to 14/05/2026 10:45:58.000)
- Event Details:**
 - LogName: Security
 - EventCode: 4732
 - EventType: 0
 - ComputerName: EC2AMAZ-CLINT01.splunkadmin.local
 - SourceName: Microsoft Windows security auditing
 - Type: Information
 - RecordNumber: 323768
 - Keywords: Audit Success
 - TaskCategory: Security Group Management
 - OpCode: Info
 - Message: A member was added to a security-enabled local group.
- Subject:**
 - Security ID: S-1-5-21-3021654831-3858487264-1625118876-1120
 - Account Name: admin
 - Account Domain: SPLUNKADMIN
 - Login ID: 0x28974c
- Member:**
 - Security ID: S-1-5-21-3021654831-3858487264-1625118876-1125
 - Account Name: -
- Group:**
 - Security ID: S-1-5-32-544
 - Group Name: Administrators
 - Group Domain: builtin
- Additional Information:**
 - Privileges: -

Note: If the Member SID is same in both the Server and Client logs, then the same Domain User was added to both Domain and Local Admin Privileges, which results in giving full control to local client machines and can access DC.

2. Extraction and Detection

1) Build Detection for 4728 (Domain Admin addition)

i) Extract Important fields from _raw for EventCode 4728

Run:

```
index=win_ad_idx EventCode=4728
| rex "(?is)Member:\s+Security ID:\s+(?<member_sid>S-[0-9\-]+)"
| rex "(?is)Member:.*?Account Name:\s+cn=(?<added_user>[^\s,]+)"
| rex "(?is)Group:.*?Group Name:\s+(?<group_name>[^\r\n]+)"
| rex "(?is)Subject:.*?Account Name:\s+(?<actor>[^\r\n]+)"
| table _time ComputerName member_sid added_user group_name actor
```

Expected:

```
member_sid => who got access
added_user => <added_domainusername>
group_name => Domain Admins
actor => <admin_who_added>
```

The screenshot shows the Splunk Enterprise Search interface. The search query is: `index=win_ad_idx EventCode=4728`. The search results are displayed in a table with the following columns: `_time`, `ComputerName`, `member_sid`, `added_user`, `group_name`, and `actor`. The results show 1 of 181 events matched.

_time	ComputerName	member_sid	added_user	group_name	actor
2020-05-14 10:22:22	EC2AMAZ-DC01.splunkadmin.local	S-1-5-21-3021654831-3858487284-1625118876-1125	Eric Wan	Domain Admins	admin

ii) Detection logic for the EventCode 4728 (Domain Admins addition)

Run:

```
index=win_ad_idx EventCode=4728
| rex "(?is)Member:\s+Security ID:\s+(?<member_sid>S-[0-9\-]+)"
| rex "(?is)Member:.*?Account
Name:\s+cn=(?<added_user>[^\,]+)"
| rex "(?is)Group:.*?Group
Name:\s+(?<group_name>[^\r\n]+)"
| rex "(?is)Subject:.*?Account
Name:\s+(?<actor>[^\r\n]+)"
| search group_name="Domain Admins"
| table _time ComputerName added_user member_sid actor
group_name
```

The screenshot shows the Splunk Search interface. The search bar contains the following query:

```
index=win_ad_idx EventCode=4728
| rex "(?is)Member:\s+Security ID:\s+(?<member_sid>S-[0-9\-]+)"
| rex "(?is)Member:.*?Account Name:\s+cn=(?<added_user>[^\,]+)"
| rex "(?is)Group:.*?Group Name:\s+(?<group_name>[^\r\n]+)"
| rex "(?is)Subject:.*?Account Name:\s+(?<actor>[^\r\n]+)"
| search group_name="Domain Admins"
| table _time ComputerName added_user member_sid actor group_name
```

The search results show 1 of 91 events matched. The results table is as follows:

_time	ComputerName	added_user	member_sid	actor	group_name
2020-05-14 18:22:22	EC2AMAZ-OC91.splunkadmin.local	Eric Man	S-1-5-21-3821654831-3858487264-1625118876-1125	admin	Domain Admins

2) Build Detection for 4732 (Local Admin addition)

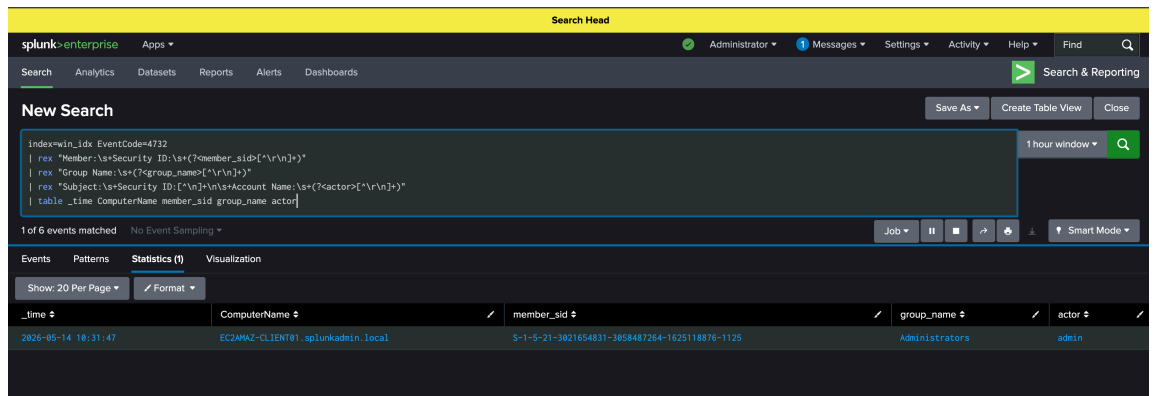
- i) Extract important fields from `_raw` for EventCode 4732

Run:

```
index=win_idx EventCode=4732
| rex "Member:\s+Security ID:\s+(?<member_sid>[^\r\n]+)"
| rex "Group Name:\s+(?<group_name>[^\r\n]+)"
| rex "Subject:\s+Security ID:[^\n]+\n\s+Account
Name:\s+(?<actor>[^\r\n]+)"
| table _time ComputerName member_sid group_name actor
```

Expected:

```
member_sid => who got access
group_name => Administrators
actor => <admin_who_added>
```



The screenshot shows the Splunk Enterprise Search interface. The search bar contains the following query:

```
index=win_idx EventCode=4732
| rex "Member:\s+Security ID:\s+(?<member_sid>[^\r\n]+)"
| rex "Group Name:\s+(?<group_name>[^\r\n]+)"
| rex "Subject:\s+Security ID:[^\n]+\n\s+Account Name:\s+(?<actor>[^\r\n]+)"
| table _time ComputerName member_sid group_name actor
```

The search results show 1 of 6 events matched. The results table is as follows:

_time	ComputerName	member_sid	group_name	actor
2026-05-14 10:31:47	EC2AWAZ-CLIENT01.splunkadmin.local	S-1-5-21-3021054831-3858487264-1625118876-1125	Administrators	admin

Note: The EventCode 4732 may or may not sometimes logs the added username. Hence, we will use the `member_sid` for the correlation detection in the later sections.

ii) Detection Logic for the EventCode 4732 (Local Admin addition)

```
Run:
index=win_idx EventCode=4732
| rex "Member:\s+Security
ID:\s+(?<member_sid>[^\r\n]+) "
| rex "Group Name:\s+(?<group_name>[^\r\n]+) "
| rex "Subject:\s+Security ID:[^\n]+\n\s+Account
Name:\s+(?<actor>[^\r\n]+) "
| search group_name="Administrators"
| stats count by _time ComputerName member_sid
actor
```

New Search

```
index=win_idx EventCode=4732
| rex "Member:\s+Security ID:\s+(?<member_sid>[^\r\n]+) "
| rex "Group Name:\s+(?<group_name>[^\r\n]+) "
| rex "Subject:\s+Security ID:[^\n]+\n\s+Account Name:\s+(?<actor>[^\r\n]+) "
| search group_name="Administrators"
| stats count by _time ComputerName member_sid actor
```

1 of 6 events matched No Event Sampling

_time	ComputerName	member_sid	actor	count
2026-05-14 10:31:47	EC2AMAZ-CL1ENT01.splunkadmin.local	5-1-5-21-3021654831-3858487264-1625118876-1125	admin	1

3) Confirmation Evidence of Privilege Escalation Event (4672)

Run:

```
(index=win_idx OR index=win_ad_idx) EventCode=4672
| rex field=Message "(?ms)Subject:\s+Security
ID:\s+[^\n]+\n\s+Account Name:\s+(?<user>[^\r\n]+) "
| eval user=trim(user)
| where NOT like(user,"%$")
| search user="ericwan"
| table _time user ComputerName EventCode
```

Confirms: the escalated account received special admin privileges during logon.

New Search

```
(index=win_idx OR index=win_ad_idx) EventCode=4672
| rex field=Message "(?ms)Subject:\s+Security ID:\s+[^\n]+\n\s+Account Name:\s+(?<user>[^\r\n]+) "
| eval user=trim(user)
| where NOT like(user,"%$")
| search user="ericwan"
| table _time user ComputerName EventCode
```

4 events (14/05/2026 11:30:50.000 to 14/05/2026 11:45:50.000) No Event Sampling

_time	user	ComputerName	EventCode
2026-05-14 11:37:44	ericwan	EC2AMAZ-DC01.splunkadmin.local	4672
2026-05-14 11:37:44	ericwan	EC2AMAZ-DC01.splunkadmin.local	4672
2026-05-14 11:38:49	ericwan	EC2AMAZ-DC01.splunkadmin.local	4672
2026-05-14 11:38:48	ericwan	EC2AMAZ-DC01.splunkadmin.local	4672

Note: EventCode 4672 can be added as an additional privileged-logon confirmation signal for higher fidelity detection but was omitted in rest of the steps because 4624 post-escalation logon correlation already demonstrates account use after privilege assignment.

4) Post-escalation Detection (4624)i) For **Windows Server** events detection,**Run:**

```
index=win_ad_idx EventCode=4624
| rex "(?is)New Logon:.*?Account Name:\s+(?<user>[^\r\n]+)"
| search user="<added_username>"
| stats count by user ComputerName _time
```

New Search

```
index=win_ad_idx EventCode=4624
| rex "(?is)New Logon:.*?Account Name:\s+(?<user>[^\r\n]+)"
| search user="ericwan"
| stats count by user ComputerName _time
```

3 events (14/05/2026 10:52:45.000 to 14/05/2026 11:07:45.000) No Event Sampling

user	ComputerName	_time	count
ericwan	EC2AMAZ-DC01.splunkadmin.local	2026-05-14 10:58:27	1
ericwan	EC2AMAZ-DC01.splunkadmin.local	2026-05-14 10:58:29	2

ii) For **Windows Client** events detection,**Run:**

```
index=win_idx EventCode=4624
| rex "(?is)New Logon:.*?Account
Name:\s+(?<user>[^\r\n]+)"
| rex "(?is)Subject:.*?Account
Name:\s+(?<subject_user>[^\r\n]+)"
| eval user=coalesce(user, subject_user)
| search user="<added_username>"
| stats count by user ComputerName _time
```

New Search

```
index=win_idx EventCode=4624
| rex "(?is)New Logon:.*?Account Name:\s+(?<user>[^\r\n]+)"
| rex "(?is)Subject:.*?Account Name:\s+(?<subject_user>[^\r\n]+)"
| eval user=coalesce(user, subject_user)
| search user="ericwan"
| stats count by user ComputerName _time
```

3 events (14/05/2026 11:35:00.000 to 14/05/2026 11:50:00.000) No Event Sampling

user	ComputerName	_time	count
ericwan	EC2AMAZ-CLIENT01.splunkadmin.local	2026-05-14 11:37:42	1
ericwan	EC2AMAZ-CLIENT01.splunkadmin.local	2026-05-14 11:37:43	2

3. Normalization of both detections (4732 and 4728)

In Search Head UI,

Run:

```
(index=win_idx EventCode=4732) OR (index=win_ad_idx
EventCode=4728)
| rex "(?is)Member:\s+Security ID:\s+(?<member_sid>S-[0-9\-
]+)"
| rex "(?is)Group:.*?Group Name:\s+(?<group_name>[^\r\n]+)"
| rex "(?is)Member:.*?Account
Name:\s+cn=(?<added_user1>[^\s,]+)"
| rex "(?is)Member:.*?Account
Name:\s+(?<added_user2>[^\r\n]+)"
| rex "(?is)Subject:.*?Account Name:\s+(?<actor>[^\r\n]+)"
| eval added_user=coalesce(added_user1, added_user2)
| eval added_user=if(added_user=="-" OR added_user=="",
null(), added_user)
| eval source=case(EventCode==4728,"DC",
EventCode==4732,"Client")
| eval system=coalesce(ComputerName, host)
| table _time member_sid group_name added_user actor source
system EventCode
```

Expected:

member _sid	added_user	group_name	source
SID	<added_username>	Domain Admins	DC
SID	(null)	Administrators	Client

The screenshot shows the Splunk Search Head interface. The search query is pasted into the 'New Search' box. The results are displayed in a table view with columns: _time, member_sid, group_name, added_user, actor, source, system, and EventCode. Two events are shown, one for EventCode 4732 and one for EventCode 4728.

_time	member_sid	group_name	added_user	actor	source	system	EventCode
2026-05-14 10:31:47	S-1-5-21-3921654831-3958487264-1625118876-1125	Administrators	admin	admin	Client	EC2AMAZ-CLIENT01.splunkadmin.local	4732
2026-05-14 10:22:22	S-1-5-21-3921654831-3958487264-1625118876-1125	Domain Admins	Eric Wan	admin	DC	EC2AMAZ-DC01.splunkadmin.local	4728

4. Correlation Logic for Privilege Escalation related Events

Run:

```
(index=win_idx EventCode=4732) OR (index=win_ad_idx
EventCode=4728)
| rex "(?is)Member:\s+Security ID:\s+(?<member_sid>S-[0-9\-
]+)"
| rex "(?is)Group:.*?Group Name:\s+(?<group_name>[^\r\n]+)"
| rex "(?is)Member:.*?Account
Name:\s+cn=(?<added_user>[^\s,]+)"
| rex "(?is)Subject:.*?Account Name:\s+(?<actor>[^\r\n]+)"
| eval source=case(EventCode==4728,"DC",
EventCode==4732,"Client")
| eval added_user=if(added_user=="-" OR isnull(added_user),
null(), added_user)
| bin _time span=30m
| stats
    values(group_name) as groups
    values(actor) as actors
    values(added_user) as users
    values(ComputerName) as systems
    values(source) as sources
by member_sid _time
| where mvcount(groups) > 1
```

The screenshot displays the Splunk Enterprise Search Head interface. The 'New Search' panel contains the correlation logic. Below the search bar, the results are shown in a table format. The table has columns for member_sid, _time, groups, actors, users, systems, and sources. Two events are listed, both for the same member_sid and occurring at the same time. The groups column shows 'Administrators' and 'Domain Admins', the actors column shows 'admin', the users column shows 'Eric Wan', the systems column shows 'EC2AMAZ-CLIENT01.splunkadmin.local' and 'EC2AMAZ-DC01.splunkadmin.local', and the sources column shows 'Client' and 'DC'.

member_sid	_time	groups	actors	users	systems	sources
S-1-5-21-3021654831-3858487264-1625118876-1125	2026-05-14 14:30:00	Administrators	admin	Eric Wan	EC2AMAZ-CLIENT01.splunkadmin.local	Client
S-1-5-21-3021654831-3858487264-1625118876-1125	2026-05-14 14:30:00	Domain Admins	admin	Eric Wan	EC2AMAZ-DC01.splunkadmin.local	DC

Next Steps

Creating and Configuring Privilege Escalation alert in Splunk Enterprise (Search Head)
 – Refer [25_Creating_Privilege_Escalation_alert_Splunk_Enterprise_V1.pdf](#).